NATIONAL UNIVERSITY OF MODERN LANGUAGE ISLAMABAD

Assignment 01

Submitted to:

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Submitted by:

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Roll No: …………………… SP21379

Subject: …………………… computer network

Department: ……………… BSSE-5-Afternoon

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Working with the OSI Model, Worksheet

the OSI layer journey when sending an email from your computer to an email server.

Layer	Name	Protocol	Up/Dn	What the layer does in this context
7	Application	Email Client(SMTP)	Down	The email client uses SMTP to send the email to the server.
6	Presentation	MIME	Down	Encodes the email content in a format suitable for transmission.
5	Session	Port 25	Down	Establishes a session with the email server using port 25 for SMTP.
4	Transport	TCP	Down	Creates a TCP packet with headers, ensuring reliable delivery.
3	Network	IP	Down	Adds IP addresses for routing the packet to the recipient's email server.
2	Data Link	Ethernet	Down	Encodes packet into Ethernet frames for LAN transmission.
1	Physical	Cat-5	Down	Transmits the email packet as electrical signals over the Ethernet cable.

the process of downloading a file from an FTP server.

Layer	Name	Protocol	Up/Dn	What the layer does in this context
7	Application	FTP	Up	FTP client initiates a request to download a file from the server.
6	Presentation	ASCII Encoding	Up	Ensures the data format is compatible for viewing once downloaded.
5	Session	Port 21	Up	Maintains the session between client and FTP server using port 21 for control.
4	Transport	TCP	Up	Ensures error-free transmission and establishes the data packet sequence.

3	Network	IP	Up	Routes the data packet to your IP address through the network.
2	Link Data	Wi-Fi (802.11)	Up	Formats data for Wi-Fi transmission and provides MAC addressing.
1	Physical	Wireless Signal	Up	Transmits data as radio waves over a wireless connection.

the journey of accessing a secure website, where data needs encryption (HTTPS).

Layer	Name	Protocol	Up/Dn	What the layer does in this context
7	Application	HTTPS	Down	The browser initiates a secure connection to the website using HTTPS.
6	Presentation	SSL/TLS	Down	Encrypts data to ensure secure transmission over the internet.
5	Session	Port 443	Down	Establishes a session with the web server over port 443 (HTTPS).
4	Transport	TCP	Down	Manages reliable data transfer and segments the data into packets.
3	Network	IP	Down	Routes the packets to the web server's IP address across different networks.
2	Data Link	Ethernet/Wi-Fi	Down	Encapsulates data into frames and adds MAC addresses for LAN transmission.
1	Physical	Fiber Optic Cable	Down	Converts data packets into light pulses for transmission over fiber optic cable.